

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA14

COURSE NAME: COMPILER DESIGN

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 25

Total Marks:50

Mode: In Class

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I_K.Madhulika(192524190) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:K. Madhulika

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization (15)	Accuracy (10)	Clarity (5)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	

6	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
7	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
8	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
9	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
10	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
11	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
12	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
13	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
14	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
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16	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
17	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
18	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
19	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
20	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
21	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
22	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
23	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
24	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
25	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
		OVERALL RUBRICS VALUE					
	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 25	/ 20	/ 15	/ 10	/ 5	/ 100

Annexure A

SHORT ANSWER TEST

- Q1. Define the role of a parser in compiler design.
- Q2. What is a context-free grammar (CFG)?
- Q3. Identify terminals and non-terminals in:
 $S \rightarrow aSb \mid \epsilon$
- Q4. Remove left recursion for:
 $A \rightarrow Aa \mid b$
- Q5. Apply left factoring for:
 $S \rightarrow abA \mid abB \mid c$
- Q6. Find FIRST(S) for:
 $S \rightarrow aA \mid b$
- Q7. Find FIRST(A) for:
 $A \rightarrow aA \mid \epsilon$
- Q8. Find FOLLOW(S) for:
 $S \rightarrow AB$
- Q9. Find FOLLOW(B) for:
 $S \rightarrow AB, B \rightarrow b$
- Q10. Check whether the grammar is ambiguous:
 $E \rightarrow E + E \mid id$
- Q11. Construct a parse tree for:
id + id using
 $E \rightarrow E + E \mid id$
- Q12. Check whether the string aaabbb is valid for:
 $S \rightarrow aSb \mid \epsilon$
- Q13. Check whether the string ab is valid for:
 $S \rightarrow aA, A \rightarrow b$
- Q14. What is top-down parsing?
- Q15. What is recursive descent parsing?
- Q16. Write the procedure for non-terminal S in:
 $S \rightarrow aS \mid b$
- Q17. What is predictive parsing?
- Q18. State the condition for a grammar to be LL(1).
- Q19. What is bottom-up parsing?
- Q20. What is shift-reduce parsing?
- Q21. List the four actions in shift-reduce parsing.
- Q22. What is operator precedence parsing?
- Q23. What is an LR parser?
- Q24. What is an SLR parser?
- Q25. What is panic mode error recovery?

Bottom of Form

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis

		analytical ability			
Organization	15%	Well-structure d answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organizatio n, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandabl e with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Name: K. Madhulika

Regno: 192524190

Course code/Slot: CSA 1419/Slot A

Course name: Compiler design

1. The parser in the compiler design comes under syntax phase. The input for the parser are the tokens and the output is the parse tree.

2. Context-free grammar (CFG) is defined by 4 actions.

$$CFG = \{V, T, P, S\}$$

$$3. S \rightarrow aSb \mid \epsilon$$

$$S \rightarrow aSb$$

$$S \rightarrow \epsilon$$

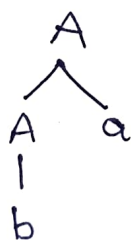
Non-terminals = a, S, b

terminals = ϵ

$$4. A \rightarrow Aa \mid b$$

$$A \rightarrow Aa$$

$$A \rightarrow b$$



= ba

5. $S \rightarrow abA \mid abB \mid c$
 $S \rightarrow abA$
 $S \rightarrow abB$
 $S \rightarrow c$

S
|
c

6. $S \rightarrow aA \mid b$
 $S \rightarrow aA$
 $S \rightarrow b$

$$\text{FIRST}(S) = \{a, b\}$$

7. $A \rightarrow aA \mid \epsilon$
 $A \rightarrow aA$
 $A \rightarrow \epsilon$

$$\text{FIRST}(A) = \{a, \epsilon\}$$

8. $S \rightarrow AB$

$$\text{FOLLOW}(S) = \{\$ \}$$

9. $S \rightarrow AB$

$$B \rightarrow b$$

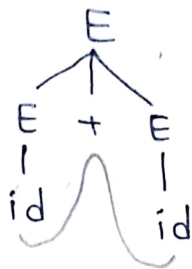
$$\text{FOLLOW}(B) = \{\$ \}$$

10. $E \rightarrow E + E \mid id$

$$E \rightarrow E + E \mid id$$

$$E \rightarrow E + E$$

$$E \rightarrow id$$

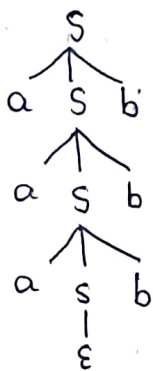


$\therefore id + id$

12.

Given, string aaabbb

$$S \rightarrow aSb \mid \epsilon$$



$= a a a \epsilon b b b$

$= a a a b b b$

$\therefore$ Given string is valid.

13.

Given, string is ab

$$S \rightarrow aA$$

$$A \rightarrow b$$



$= ab$

$\therefore$ Given string is valid.

14. Top-down parsing:

We solve the productions from parent node to get leaf node. This type of parsing is called top-down parsing.

Top down parsing is classified into 2 types predictive parsing and recursive descent parsing.

19. Bottom-up parsing:

We solve the productions from leaf node to get parent node. This type of parsing is called bottom-up parsing. Shift-reducer parsing is a type of bottom-up parsing.

20. Shift-reduce parsing:

Productions and input string will be given.

Three columns will be present stack, input and action. We have to shift the input string from input to stack and reduce them based on the productions given. Since, it is a type of bottom up parsing we will get the final production using input string.

21. Four actions in shift-reduce parsing

1. Shift

2. Reduce

3. Accept

4. Reject